

Bigyan Aryal

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EDUCATION

Bachelor of Engineering (B.E) - Computer Engineering

2023 – 2027

Paschimanchal Campus, Institute of Engineering, Tribuvan University
Pokhara, Nepal

SKILLS

Languages: Python, C, C++

Libraries: PyTorch, NumPy, Scikit-learn, Pandas, Open-CV, FastAPI, Seaborn, Matplotlib

Machine learning: Feature Engineering, Optimization, EDA, Natural Language Processing, Computer Vision.

Tools: Git, Github, VS Code, Jupyter Notebook, Conda, Docker

Soft Skills: Problem-Solving, Communication, Mass Presentation, Team Collaboration.

RELEVANT EXPERIENCE

Innovative Computer Engineering Student's Society

December 2025 – February 2026

ML Mentor

- Designed and delivered a hands-on ML curriculum for 10+ junior students covering the full pipeline: **data cleaning**, **EDA**, **feature engineering**, and **model building** using **NumPy**, **Pandas**, **Scikit-learn**, **Matplotlib**, and **Seaborn**.
- Guided 10+ students to complete **end-to-end ML** project by the end of the mentorship, and enabled them to successfully present their work and **demo** to the live audience.

PROJECTS

Image Captioning Model [Github repository link](#)

- Fusion of **Computer Vision** and **Natural Language Processing**, It uses **CNN (Resnet50)** as encoder and **RNN** as decoder.
- Wrapped the model using **FastAPI** and containerized the model using **Docker** then deployed the container to **Azure Container Registry**.

MeroPDF: RAG-based PDF chatbot [Github repository link](#)

- Architected a full-stack **RAG** system (Next.js + FastAPI + Qdrant + MongoDB + Azure Blob Storage) with multi-user session isolation, JWT authentication, and persistent chat history.
- Evaluated embedding models using **MTEB benchmarks**, selecting **BAAI/bge-base-en-v1.5** (retrieval score: **54.75**), and assessed pipeline quality via **RAGAS** while achieving Context Recall of **0.87** on a 25-question golden dataset.
- Analyzed system limitations critically, identifying a faithfulness gap (**RAGAS: 0.44**) and proposing concrete fixes and demonstrating research-level thinking beyond just shipping the project.

Shakespeare GPT: Transformer Implementation [Github repository link](#)

- Improved understanding of LLM internals, by successful training convergence, and implementing attention, positional encoding, and transformer blocks from scratch in PyTorch.
- Generated coherent text outputs by training a tiny Transformer model on the Tiny Shakespeare dataset and iterating on training parameters.

AWARDS AND ACHIEVEMENTS

• National Level 48 hour Hackathon: VectorHack 2026, Winner

Awarded for the project '**NeuroRead**', A mobile app for **dyslexic** people to improve their ability of **phonemic** awareness. Uses **finetuned BART** model for dyslexic friendly **summarization** of study materials.

• OpenStreetMap(OSM) Hackfest, Runner Up (2026)

Awarded for the **NightTime Route Planner Mobile App**, Integrated with ML model that highlighted safer areas based on the **OpenStreetMap** data combined with synthetically generated data